

A note on “a fully dynamic multi-secret sharing scheme with redundant authorization”

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Abstract. We show that the secret sharing scheme [Cryptogr. Commun. 16(1): 3-20 (2024)] cannot be put into practice. (1) It confused the elements in a residue class ring modulo a prime p with the points in an elliptic curve group over the finite field F_p . (2) It confused the underlying elliptic curve with the Lagrange interpolating curve, and falsely requires the interpolating polynomial to map a point on the elliptic curve to another point on the same elliptic curve. (3) It misuses the bit-wise XOR operator for the operands with unequal bit-length, which results in the exposure of any participant’s share, and the loss of confidentiality.

Keywords: Secret sharing, Residue class ring, Elliptic curve group, Lagrange interpolating polynomial, Confidentiality.

1 Introduction

Key management is the set of techniques and procedures, which supports initialization of system users within a domain, generation, distribution, installation, update, revocation, destruction of keying material [3]. The motivation for secret sharing is to safeguard cryptographic keys from loss, by dividing a secret into pieces called shares which are distributed amongst users such that the pooled shares of specific subsets of users allow reconstruction of the original secret. Shamir’s (t, n) threshold secret sharing scheme [4] is a method by which a trusted party computes secret shares $S_i, 1 \leq i \leq n$, from a secret S , and securely distributes S_i to user P_i , such that any t or more users who pool their shares may recover S , but any group knowing only $t - 1$ or fewer shares may not.

Recently, Li et al. [2] presented a multi-secret sharing scheme based on the intractability of Elliptic Curve Discrete Logarithm Problem (ECDLP) and that of Shamir’s (t, n) threshold secret sharing scheme. We find the Li et al.’s scheme cannot be practically implemented. It confused the elements in a residue class ring modulo a prime p with the points in an elliptic curve group over the finite field F_p . In its verification equation, the Lagrange interpolating polynomial $E_1(x)$ is reconstructed by using pairs (t_i, x_i) , where $t_i, x_i \in Z_p \times Z_p$ are points on the underlying elliptic curve. But the polynomial cannot be used to map a point t_i on an elliptic curve to the point x_i on the same elliptic curve. It confused the elliptic curve with the interpolating curve. The bit-wise XOR operator is misused for the verification equation, in which the operands have unequal bit-length. The party P_u ’s share t_u is exposed, which will result in the loss of confidentiality.

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2 Review of the Li et al's secret sharing scheme

The secret sharing scheme [2] consists of five phases: Initialization, Construction, Verification, Recovery, and Dynamic update. Some symbols and their descriptions are listed below (see Table 1). The scheme can be restated as follows. For readers' conveniences, we colored some flawed computations.

Table 1: Symbols and descriptions

symbol	description	symbol	description
n	number of participants	k	threshold
D	dealer	P_i	i -th participant with identity id_i
H	hash function	S_i	i -th shared secret
E	elliptic curve	G	elliptic curve group with generator S

2.1 Initialization

- Step-1. D selects an elliptic curve E over the finite field F_p where p is a big prime. Select the elliptic curve group G with a generator S , and a hash function H .
- Step-2. Each participant $P_i, i = 1, 2, \dots, n$ picks $r_i \in Z_p^*$ as his own share and computes $R_i = r_i \cdot S$, $B_i = r'_i \cdot S$, where $r'_i = R_i - r_i$. Then, P_i saves r_i and sends (R_i, B_i) to D .
- Step-3. D checks the uniqueness of R_i and $B_i, i = 1, 2, \dots, n$.
- Step-4. D publishes (id_i, R_i) and $E(x)$, where $id_i = i, i = 1, 2, \dots, n$, and $E(x)$ is a polynomial constructed by using the n pairs (id_i, B_i) . id_i is the unique identity information of the participant P_i .

2.2 Construction

- Step-1. D picks $j \in Z_p^*$ as his private key, and publishes $R = j \cdot S$ and $T = H(S_1 || S_2 ||)$.
- Step-2. D computes $t_i = j \cdot R_i, i = 1, 2, \dots, n$, where t_i is recorded as the pseudo-share of the participant P_i .
- Step-3. D computes $T_i = M_i \oplus t_i \oplus C$, $x_i = M_i \cdot S, i = 1, 2, \dots, n$, where $M_i, C \in Z_p^*$ ensuring that $x_1, x_2, \dots, x_n, R, R_1, R_2, \dots, R_n, B_1, B_2, \dots, B_n$ are different with each other, and M_i, t_i are also different with each other. Then, D saves C and publishes $T_i, i = 1, 2, \dots, n$, where C is represented as the redundancy.
- Step-4. D constructs a $k-1$ degree polynomial $g(x)$ by using S_1, C and picked $a_1, \dots, a_{k-1} \in Z_p, a_{k-1} \neq 0$, such that

$$g(x) = (S_1 + C) + a_1x + a_2x^2 + \dots + a_{k-1}x^{k-1}$$

- Step-5. D computes $v_i = (x_i, y_i)$, where $y_i = g(x_i), i = 1, 2, \dots, n$.

- Step-6. D publishes polynomials $E_1(x)$ and $E_2(x)$, where $E_1(x)$ is constructed by using the pairs $v'_i = (t_i, x_i)$ and $E_2(x)$ is constructed by using the pairs $v''_i = (t_i, y_i)$, $i = 1, 2, \dots, n$.
- Steps 7-10. See the original description (pages 8-9, [2]).

2.3 Verification

Assume that the pseudo-share provided by P_u is t_u^* . P_a checks that

$$[(T_u \oplus C) \oplus t_u^*] \cdot S = x_u = E_1(t_u^*)$$

If it holds, P_a determines that P_u is honest, i.e., $t_u^* = t_u$.

For Recovery phase and Dynamic update phase, we refer to the original descriptions (pages 9-12, [2]).

3 Cryptanalysis of the secret sharing scheme

Though the Li et al's secret sharing scheme is interesting, we find there are some inconsistent computations. Besides, it misused the bit-wise XOR operator.

3.1 Inconsistent computations

Let E be an elliptic curve defined over a field. There is a chord-and-tangent rule for adding two points to give a third point [1]. Concretely, let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be two distinct points on an elliptic curve E . Then the sum R , of P and Q , is defined as follows. First draw a line through P and Q ; this line intersects the elliptic curve at a third point. Then R is the reflection of this point about the x -axis.

The double R , of P , is defined as follows. First draw the tangent line to the elliptic curve at P . This line intersects the elliptic curve at a second point. Then R is the reflection of this point about the x -axis.

Scalar multiplication is represented as $k \cdot P$, where k is an integer and P is a point on the elliptic curve, i.e.,

$$k \cdot P = \underbrace{P + P + \dots + P}_{k \text{ times}}$$

which dominates the execution time of elliptic curve cryptographic schemes.

We find the Li et al's scheme has confused the structure of elliptic curve groups. As we see, the participant P_i needs to pick $r_i \in Z_p^*$ and compute

$$R_i = r_i \cdot S, \quad r'_i = R_i - r_i, \quad B_i = r'_i \cdot S$$

where S is a generator of the elliptic curve group G . Notice that $R_i \in G$ is a point on the underlying elliptic curve, while r_i is an integer. The computation $r'_i = R_i - r_i$ makes no sense because R_i is a vector and r_i is a scalar. Furthermore,

$$B_i = r'_i \cdot S = (R_i - r_i) \cdot S = R_i \cdot S - r_i \cdot S$$

where $R_i \cdot S$ represents the multiplication of two points on the elliptic curve, not the usual scalar multiplication. But as so far, such computations are not defined in elliptic curve cryptography. Frankly speaking, the scheme has confused the elements in a residue class ring modulo a prime p with the points in an elliptic curve group over the finite field F_p .

3.2 Failure of reconstructing the interpolating polynomials

The famous Lagrange interpolation method says that: given k pairs $(x_i, y_i) \in \mathbb{R} \times \mathbb{R}$, $i = 1, 2, \dots, k$, satisfying that $y_i = f(x_i)$ where $f = f_0 + f_1x + f_2x^2 + \dots + f_{k-1}x^{k-1}$, $f_0, f_1, \dots, f_{k-1} \in \mathbb{R}$, is a polynomial of $k-1$ degree, the polynomial can be efficiently reconstructed by the formula

$$f(x) = \sum_{i=1}^k \left(y_i \times \prod_{j=1, j \neq i}^k \frac{x - x_j}{x_i - x_j} \right)$$

Notice that *both x_i and y_i are scalars, not vectors*. Otherwise, the division $\frac{x-x_i}{x_i-x_j}$ of two vectors makes no sense. But we find the scheme has neglected the basic requirement.

For example, the scheme specifies that

$$j \in Z_p^*, \quad t_i = j \cdot R_i, \quad M_i \in Z_p^*, \quad x_i = M_i \cdot S$$

and $E_1(x)$ is constructed by using the pairs $v'_i = (t_i, x_i)$, $i = 1, 2, \dots, n$. Actually, $t_i = j \cdot R_i = j \cdot r_i S \in G$. So does x_i . That means both t_i and x_i are points on the underlying elliptic curve, and represented as vectors, not scalars. In this case,

$$(t_i, x_i) \in (Z_p \times Z_p, Z_p \times Z_p), \quad i = 1, 2, \dots, n$$

The above n pairs cannot be used for reconstructing a polynomial. Likewise, The polynomial $E(x)$ cannot be constructed by using the n pairs (id_i, B_i) . So does the polynomial $E_2(x)$. Actually, the scheme confuses the underlying elliptic curve with the Lagrange interpolating curve, and falsely requires the interpolating polynomial to map a point on the elliptic curve to another point on the same elliptic curve.

3.3 Loss of confidentiality

It is well known that the bit-wise XOR operator is broadly used in symmetric encryption algorithms, such as AES. To securely blind a target operand, both two operands should be of equal bit-length. Otherwise, a substring of the longer operand will be exposed. In the Li et al.'s scheme, the dealer needs to compute

$$T_i = M_i \oplus t_i \oplus C$$

where $M_i, C \in Z_p$ are scalars. But $t_i \in Z_p \times Z_p$ is a point over the elliptic curve. In this case, the x -coordinate (aligned from the right) or y -coordinate (aligned from the left) of t_i is almost copied into T_i , which will be published by the dealer. The adversary who has captured the public parameter T_i , can successfully retrieve the coordinate so as to recover the point t_i using the elliptic curve equation. With the share t_i , the adversary can cheat other participants. That

means the scheme fails to keep the confidentiality.

4 Conclusion

We show that the Li et al's secret sharing scheme is flawed. It cannot be practically implemented due to some inconsistent computations. It cannot keep the confidentiality owing to the exposure of shares. We hope the findings in this note could be helpful for the future work on designing such schemes.

References

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